

WasteVision: AI-Powered Waste Recognition Using YOLOv8 and OpenCV

1. Introduction:

Improper waste management has been a growing concern to our environment and its effects can be felt through significant pollution that harms our planet. This project, “WasteVision” utilises computer vision and YOLOv8 to classify and recognise different types of waste in real-time.

2. Methodology:

Data Collection –

A dataset is chosen from the internet, specifically from “Roboflow”

Model Training –

Using YOLOv8 as a model, the dataset taken from roboflow is used. The training process took multiple iterations to optimise its accuracy.

Model implementation –

The trained model is integrated in the program using OpenCV to process video input. A webcam whether built-in or external is used to capture images, which is passed to the model for processing. These are displayed via GUI made using Tkinter.

User Interface –

GUI is implemented using Tkinter, of which displays the following:

- Video feed (Built-in or External)
- Real-time classifications and results
- Buttons for switching webcam, uploading pictures

3. Results and Analysis:

Current model use achieved good detection and recognition. However, it may sometimes misclassify waste depending on their current state, distance from camera, lighting, or similarities to other waste. This may be due to the model not yet reaching its final number of epochs as the intended is 150 but so far, it has only reached below 100. With that said, an expanded dataset with more diverse images can significantly help in improving the classification accuracy of the program.

4. Conclusion:

The project has successfully implemented computer vision for waste detection and classification with the use of YOLOv8 and OpenCV. The GUI made for is provides a simple yet interactive way to visualise waste recognition. The creation of this project may very well

promote environmental awareness to people that may use this. Future improvements include expanding the dataset and optimizing model efficiency.